

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Bhavya PS(1925524017) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”
3. “Security Attacks, Threats, Risks and Their Impact on Security Goals”
4. “Steganography vs Cryptography: Hidden Security in the Digital World”
5. “Role of Number Theory in Cryptography: From Euclid to Euler”

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

From Packet Storms to Infrastructure Siege: The History of DoS & DDoS

CSA5112 – CRYPTOGRAPHY AND NETWORK SECURITY

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Denial of Service (DoS) & Distributed Denial of Service (DDoS) Attacks

What is a DoS Attack?

- A Denial of Service (DoS) attack overwhelms a website, app, or server so legitimate users can't access it.
- Attackers send large volumes of fake or malformed requests to exhaust CPU, memory, or network capacity.
- Impact: downtime, lost revenue, reputational damage, and diverted response resources.
- Aim Extortion ,Revenge or Protest, Distractions



How a DoS Attack Works

- The attacker chooses a target website or server.
- Thousands of fake requests are sent.
- The server uses its resources to process those requests.
- The server becomes overloaded.
- Genuine users experience slow loading or cannot access the website.

Example

A student portal crashes during exam results because students login with the personal email id but not with the email id the is provided to them by the university.

What is a DDoS Attack?

Definition

A **Distributed Denial of Service (DDoS)** attack is similar to a DoS attack, but instead of one computer, **thousands or even millions of infected devices** attack the target simultaneously.

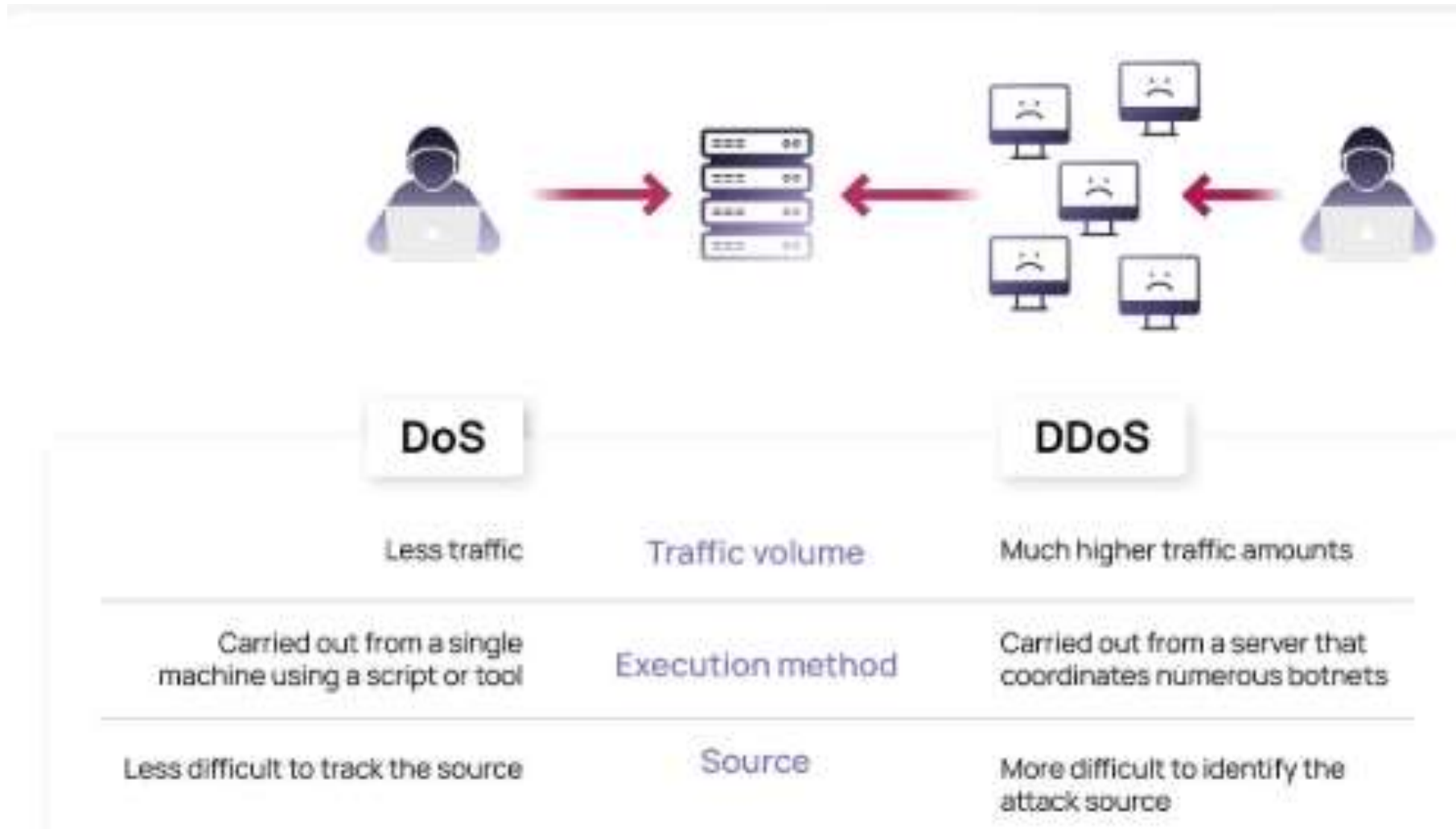
Botnet

- A collection of infected computers, phones, or IoT devices.
- Controlled remotely by the attacker.
- Every infected device sends requests to the target.

Why is DDoS More Dangerous?

- Traffic comes from many different locations.
- Harder to identify and block.
- Can affect even large organizations.

DoS vs DDoS Comparison



Real-Life Analogy

Shopping Mall Example

Normal Situation

- Customers enter normally.
- Everyone gets service.



DoS Attack

- One person repeatedly blocks the entrance.

DDoS Attack

- Thousands of fake customers crowd every entrance.
- Real customers cannot enter.

Real Examples

Include examples such as:

- GitHub DDoS attack (2018)
- Dyn DNS attack (2016)
- Banking website outages

Lessons: IoT hygiene, layered mitigation, and partnerships with transit/CDN vendors are essential.



Protection Against DoS & DDoS

- Firewall
- Web Application Firewall (WAF)
- CDN
- Load Balancer
- Rate Limiting
- CAPTCHA
- Traffic Monitoring
- AI-based Threat Detection